

L8 Control Quiz ANS

1. What control commands does the controller output to follow the trajectory from motion planning?

- A) Brake, acceleration, and steering commands
- B) Only a map-level route
- C) Object-detection bounding boxes
- D) HD-map updates

ANS: A) Brake, acceleration, and steering commands

2. In feedback control, what is the tracking error $e(t)$?

- A) The sum of reference and output
- B) The reference signal minus the system output
- C) The derivative of the reference only
- D) The control input minus the system state

ANS: B) The reference signal minus the system output

3. Which PID term uses the history of the tracking error and helps eliminate steady-state error?

- A) Proportional term
- B) Derivative term
- C) Integral term
- D) Prediction term

ANS: C) Integral term

4. What is a typical effect of increasing the derivative gain K_d ?

- A) A large increase in steady-state error
 - B) Elimination of all measurement noise
 - C) Increased overshoot
 - D) Reduced overshoot with greater sensitivity to measurement noise
- ANS: D) Reduced overshoot with greater sensitivity to measurement noise

5. What does rise time measure in a step response?

- A) The time for the transient response to rise close to the steady-state value
- B) The time at which maximum overshoot occurs
- C) The time required to compute the control law
- D) The final difference between input and output

ANS: A) The time for the transient response to rise close to the steady-state value

6. What is maximum overshoot?

- A) The time after which the output remains in a specified band
- B) The amount by which the maximum transient response exceeds the steady-state value
- C) The difference between two controller gains
- D) The minimum value reached by the reference signal

ANS: B) The amount by which the maximum transient response exceeds the steady-state value

7. What is the main characteristic of the proportional PID term?

- A) It depends only on accumulated past error
- B) It predicts future states over a horizon
- C) Its contribution is proportional to the current tracking error
- D) It is independent of the tracking error

ANS: C) Its contribution is proportional to the current tracking error

8. According to the slides, increasing K_p generally produces which combination?

- A) Slower response and larger steady-state error
- B) No change in response
- C) Lower overshoot in all cases
- D) Faster response, potentially more overshoot and oscillation

ANS: D) Faster response, potentially more overshoot and oscillation

9. What alternative name is used for Model Predictive Control?

- A) Receding Horizon Control
- B) Open-Loop Integral Control
- C) Static Gain Scheduling
- D) Randomized Feedback Control

ANS: A) Receding Horizon Control

10. At each MPC time step, how much of the optimized control sequence is normally applied?

- A) The entire sequence without further optimization
- B) Only the first control command
- C) Only the last control command
- D) No control command until the horizon ends

ANS: B) Only the first control command

11. Why did MPC decelerate earlier than PID in the sharp-turn example?

- A) MPC ignored the reference trajectory
- B) MPC used no vehicle model
- C) MPC used a look-ahead window to anticipate the turn
- D) MPC disabled longitudinal control

ANS: C) MPC used a look-ahead window to anticipate the turn

12. What is one disadvantage of MPC identified in the slides?

- A) It cannot be applied to nonlinear systems
- B) It cannot handle constraints
- C) It never uses optimization
- D) It requires an accurate, computationally efficient model and may be expensive to solve

ANS: D) It requires an accurate, computationally efficient model and may be expensive to solve

13. What is the effect of choosing a larger MPC look-ahead horizon T?

- A) Potentially better control performance but higher computation overhead
- B) Guaranteed zero computation cost
- C) More myopic behavior and less planning
- D) Elimination of the system model

ANS: A) Potentially better control performance but higher computation overhead

14. What distinguishes dynamics from kinematics?

- A) Dynamics ignores mass and forces
- B) Dynamics considers forces, mass, and inertia that affect motion
- C) Dynamics uses only geometric relationships
- D) Dynamics cannot describe vehicle motion

ANS: B) Dynamics considers forces, mass, and inertia that affect motion

15. What is the state vector of the kinematic bicycle model presented in the slides?

- A) Steering, throttle, and brake
- B) Acceleration, force, and torque
- C) Position x, position y, and heading angle theta
- D) Wheel speed, tire pressure, and yaw torque

ANS: C) Position x, position y, and heading angle theta

16. Which pair represents the control inputs of the presented kinematic bicycle model?

- A) Mass and inertia
- B) Position and heading
- C) Longitudinal and lateral forces
- D) Steering angle and vehicle speed

ANS: D) Steering angle and vehicle speed

17. For the kinematic bicycle model, how is the turning radius R related to wheelbase L and steering angle delta?

- A) $R = L / \tan(\delta)$

B) $R = L \tan(\delta)$

C) $R = \delta / L$

D) $R = vL$

ANS: A) $R = L / \tan(\delta)$

18. What happens to the turning radius as the steering angle approaches zero?

A) It approaches zero

B) It approaches infinity, corresponding to straight-line motion

C) It becomes equal to vehicle speed

D) It becomes negative regardless of direction

ANS: B) It approaches infinity, corresponding to straight-line motion

19. In the clockwise racetrack convention used in the slides, what does a positive cross-track error indicate?

A) The car is exactly on the desired trajectory

B) The car is inside the track region

C) The car is outside the track region

D) The car has stopped

ANS: C) The car is outside the track region

20. How does Twiddle tune PID gains?

A) It requires an exact nonlinear vehicle model

B) It changes all gains randomly without evaluation

C) It applies pole placement to a linear model

D) It systematically adjusts each gain and its step size according to whether the error improves

ANS: D) It systematically adjusts each gain and its step size according to whether the error improves